

MIRAGE KITTEN NIGHTLEDGER DLL SIDELOADING

Behavioral detection for AppVShNotify
loading a co-located SspiCli module outside
the Windows directory.

PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceImageLoadEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

04 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceImageLoadEvents

ATT&CK

T1574.001

Evidence-led dossier. Query status remains an untested implementation
sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Detect the process-module relationship, not a campaign hash.

Kaspersky observed NightLedger masquerading as SspiCli.dll and abusing a legitimate AppVShNotify.exe through DLL search-order hijacking. Co-location outside the Windows directory is directly observable and materially narrows benign overlap.

OPERATIONAL VALUE

Durable beyond the published sample.

The logic does not require the mutex, C2 domain or file hash. It detects the execution relationship needed by this sideload path while retaining paths and hashes for immediate investigation.

8.8

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Candidate

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- NightLedger masquerades as SspiCli.dll.
- A legitimate AppVShNotify.exe is used for DLL search-order hijacking.
- The malicious DLL is co-located with the executable and forwards expected exports.

ANALYTICAL ASSESSMENT

What the evidence supports

- The process-module pair plus co-location is more specific than either filename alone.
- Excluding the Windows directory reduces operating-system and servicing overlap.
- A match supports sideloading triage but does not by itself attribute Mirage Kitten.

DETECTION ASSUMPTIONS

What must be validated

- DeviceImageLoadEvents records the module load in the tenant.
- FolderPath and InitiatingProcessFolderPath use comparable normalized semantics.
- Legitimate packaging does not commonly co-locate this pair outside C:\Windows.

Confidence boundary

The non-Windows path condition is analyst-authored tuning, not a source fact. Promotion requires a controlled replay and a historical prevalence review.

02

THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Stage trusted binary

A legitimate AppVShNotify executable is placed in an attacker-controlled directory.

02

Co-locate DLL

A malicious file named SspiCli.dll is stored beside the binary.

03

Start process

AppVShNotify executes from the non-Windows location.

04

Delay-load module

The process loads the co-located SspiCli.dll.

05

Backdoor runs

NightLedge functionality executes in the trusted process context.

Observable core

One DLL-load event tying AppVShNotify.exe to a co-located SspiCli.dll, with both paths outside the Windows directory.

Non-observable boundary

DLL contents, successful export forwarding, mutex state, C2 response, operator commands and attribution.

DETECTION HYPOTHESIS

A repeated sideload should create a precise module-load event.

If an attacker repeats this NightLedge sideload path, DeviceImageLoadEvents should show AppVShNotify.exe loading SspiCli.dll from the same non-Windows directory. Legitimate activity can be separated through path ownership, binary signature/hash, software inventory and controlled baseline evidence.

REQUIRED TELEMETRY

DeviceImageLoadEvents

- Microsoft Defender for Endpoint DLL-load events.
- FileName and FolderPath for the loaded module.
- InitiatingProcessFileName and InitiatingProcessFolderPath.

OPTIONAL ENRICHMENT

Context that improves disposition

- SHA1 and signer reputation for both files.
- DeviceFileEvents for placement chronology.
- DeviceProcessEvents and DeviceNetworkEvents for follow-on behavior.

UNAVAILABLE HERE

Do not infer

- DLL content and export behavior.
- Mutex and encrypted C2 payload.
- Campaign attribution from this event alone.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceImageLoadEvents

Contains DLL-load events with module, initiating process, path, account, hash, device and event identifiers.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The table can prove that a module was loaded by a process. It cannot prove the loaded bytes are malicious without file inspection or corroborating behavior.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Module event	DeviceImageLoadEvents	Primary DLL-load evidence.	REQUIRED
Loaded module	FileName, FolderPath	Name and location of SspiCli.dll.	REQUIRED
Loader	InitiatingProcessFileName	AppVShNotify identity.	REQUIRED
Loader path	InitiatingProcessFolderPath	Co-location and Windows-directory test.	REQUIRED
Device	DeviceId, DeviceName	Endpoint pivot.	REQUIRED
Event identity	Timestamp, ReportId	Unique-event investigation.	REQUIRED
Hashes	SHA1, InitiatingProcessSHA1	Sample and prevalence enrichment.	OPTIONAL

Data quality requirements

- Confirm DLL-load telemetry is enabled and retained.
- Validate path semantics with a sample event.
- Preserve raw events and both hashes.
- Inventory legitimate App-V or packaging workflows.

Known failure conditions

- Module-load collection is unavailable.
- Attacker uses a different host binary or DLL name.
- Paths are absent or normalized incompatibly.
- Attacker stages the pair inside an excluded Windows path.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

PRODUCTION CANDIDATE

DETECTION OBJECTIVE

Find the source-observed sideload relationship.

Select only AppVShNotify loading a co-located SspiCli module outside the Windows directory.

ANALYTICAL LOGIC

Direct filters and investigable output.

Four direct predicates; no IOC, join, sequence, threshold, aggregation, rarity or automatic attribution.

KQL · UNTESTED CANDIDATE

```
// Title: AppVShNotify loads co-located SspiCli outside the Windows directory
// Platform: Microsoft Defender XDR Advanced Hunting
// Status: Production candidate - Untested implementation sketch
// Confidence: Medium
// Telemetry prerequisite: Microsoft Defender for Endpoint; DeviceImageLoadEvents
// Evidence: Kaspersky observed NightLedger masquerading as SspiCli.dll and using
// AppVShNotify.exe for DLL search-order hijacking. The malicious DLL is co-located
// with the legitimate executable. The non-Windows-path restriction is an analyst
// assumption intended to reduce servicing and operating-system overlap.
DeviceImageLoadEvents
| where InitiatingProcessFileName =~ "AppVShNotify.exe"
| where FileName =~ "SspiCli.dll"
| where FolderPath =~ InitiatingProcessFolderPath
| where not(InitiatingProcessFolderPath startswith @"C:\Windows\")
| project
    Timestamp,
    DeviceId,
    DeviceName,
    InitiatingProcessAccountName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessSHA1,
    FolderPath,
    FileName,
    SHA1,
    ReportId
| order by Timestamp desc
```

REVIEW FIELDS · BASELINE LOCALLY · VALIDATE BEFORE PRODUCTION

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested candidate	None	None	DeviceName + Timestamp + ReportId	Manual triage

What a match means

The documented host binary loaded a same-directory SspiCli.dll from a non-Windows path and requires immediate file and process validation.

What a match does not prove

Malicious DLL contents, successful C2, operator activity, data theft or Mirage Kitten attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify both filenames, paths, hashes, device and event identifiers.
- 02

Validate binary trust.
Confirm the AppVShNotify signature, source and expected installation path.
- 03

Collect the DLL.
Review signer, hash, prevalence, PE exports and acquisition time.
- 04

Build the timeline.
Pivot to file creation, process start, child processes and network events.
- 05

Establish ownership.
Check software distribution, App-V packaging and change records.

Escalation gate
Escalate when the pair is unowned, newly staged, unsigned or hash-unknown and followed by suspicious process, file or network activity.

Primary benign overlap
Software packaging, App-V servicing, forensic labs or security tests could stage legitimate copies outside Windows. Tune only with verified path, hash, signer and owner combinations.

INVESTIGATION PIVOTS

Host
Coverage, role and neighboring alerts.

Loader
Signature, hash, path and prevalence.

Module
Hash, exports, signer and origin.

Files
Creation source and co-location timing.

Process
Parent, command line and descendants.

Network
Connections from the loader after module load.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Application packaging or servicing	May copy Microsoft binaries and dependencies into a staging directory.	Require approved package, signer/hash and change record.
Security testing	May reproduce a DLL search-order hijack.	Scope to authorized device, user and test window.
Forensic or malware lab	May execute samples from nonstandard directories.	Allowlist isolated lab assets, never filename alone.

Baseline plan

- Run across 30-90 days.
- Inspect every match and raw event.
- Classify paths by owner and package.
- Measure field completeness before alerting.

Validation procedure

- Generate a benign same-directory load in an isolated host.
- Confirm FolderPath equality semantics.
- Verify System32 activity is excluded.
- Review all historical matches for stable benign overlap.

TEST CASES

CASE	EXPECTED RESULT
Controlled safe sideload pair outside Windows	Expected match with complete paths and hashes.
Normal System32 execution	No match because loader path begins with C:\Windows.
AppVShNotify loads another DLL	No match.
Another process loads SspiCli.dll	No match.

Hunting Initial query	Candidate Current status	Production Measured precision	Optimized Stable feedback
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Success

Controlled replay matches, normal Windows execution does not, and residual matches are low-volume and explainable.

Rejection

Path semantics cannot establish co-location or legitimate packages frequently reproduce the pair without a stable discriminator.

Failure

DeviceImageLoadEvents is absent or omits the required paths.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1574.001	Hijack Execution Flow: DLL	Directly covers loading a maliciously named DLL through a legitimate executable search-order path.

Detection coverage summary

Strong for this AppVShNotify/SspiCli relationship outside Windows; weak for renamed modules, alternative host binaries or events hidden by telemetry gaps.

Residual risk

Attackers can change the loader, DLL name or staging directory, while exact module content and attribution remain outside the event.

ANALYST CONCLUSION

A defensible behavioral candidate with a narrow validation burden.

The source-observed process-module relationship and documented Defender schema support a Production candidate. It remains an untested implementation sketch until a controlled replay verifies path semantics and the tenant baseline confirms manageable benign overlap.

REFERENCES

- 1. Kaspersky GReAT, “Mirage Kitten targets Middle East and Africa region with new malware,” 28 July 2026. <https://securelist.com/mirage-kitten-new-tools/120811/>
- 2. Microsoft Learn, “DeviceImageLoadEvents table,” accessed 4 August 2026. <https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceimageloadevents-table>
- 3. MITRE ATT&CK, T1574.001. <https://attack.mitre.org/techniques/T1574/001/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	04 August 2026	Candidate	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.